mc6801-tools User's Manual

Assembler, linker, archiver and utilities for the Motorola MC6801

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Chapter 1

Introduction

mc6801-tools is a set of Unix-style development tools for the Motorola MC6801 and MC6803 microprocessors, and for machines built around them, such as the Tandy MC-10. They run on Linux and other Unix-like systems.

Tool	Purpose
`as01`	Assembler: turns MC6801 assembly source into relocatable object modules or absolute binaries
`ld01`	Linker: combines object modules and libraries into an executable
`ar01`	Archiver: collects object modules into libraries
`objdump01`	Displays the contents of executables; also installed as `nm01` and `size01`
`objcopy01`	Converts executables into S-records, Intel hex, raw binaries and cassette images

1.1 Why a linker?

Many 6800-family assemblers are *absolute* assemblers: a program is one source file (perhaps with include files), assembled at a fixed address straight into a loadable image. `as01` can work that way too.

Its main strength, though, is *separate assembly*. You can split a program into modules, each assembled on its own into an object file. Modules export the symbols they define and import the ones they use. The linker then joins them, resolves the references between them and places the whole program at its final address. Commonly used modules can be kept in libraries, and the linker only includes the ones that a program actually needs.

1.2 Typical workflow

```
prog.s --as01--> prog.o --+
util.s --as01--> util.o --+--ld01--> prog.aout --objcopy01--> .wav .s19 .hex .bin
```

```
a.o b.o --ar01--> libx.a -+              |
                          +--objdump01--> symbols, sizes, dump
```

1.3 Heritage

The tools come from the MC6809 configuration of the assembler, linker and archiver in Bruce Evans' C compiler (`bcc`), as maintained in the *dev86* package for Linux (https://github.com/lkundrak/dev86). They have been adapted for the MC6801 instruction set.

The assembler syntax follows the one described in section C.5 of the *Introl-C Compiler Reference Manual*. The object file format is the *Introl Linkable Binary File Format*, described in the *Introl Linker and Loader Reference Manual*. Both are available at http://www.swtpc.com/mholley/Introl/.

Chapter 2

Building and Installing

2.1 Requirements

You need a C compiler and `make`. Building this manual also needs `pandoc` and a LaTeX installation (see Section 2.5).

2.2 Building

From the top of the source tree:

```
$ make
```

This builds `ar/ar01`, `as/as01`, `ld/ld01`, `ld/objdump01` and `ld/objcopy01`. The build stops at the first error.

The version number, which `as01 -v` and `ld01 -v` print, is taken from the `VERSION` file at the top of the source tree.

2.3 Installing

```
$ sudo make install
```

All the programs are installed into a single directory, by default `/usr/local/m6801-tools`. `nm01` and `size01` are created there as links to `objdump01`. The manual pages are installed into `/usr/local/share/man/man1`. To choose other directories, set `PREFIX` for the programs and `MANDIR` for the manual pages; to stage the files for packaging, set `DESTDIR` as well:

```
$ make install PREFIX=/opt/m6801
$ make install DESTDIR=/tmp/pkgroot PREFIX=/usr/bin MANDIR=/usr/share/man
```

Add the installation directory to your `PATH`:

```
$ export PATH=$PATH:/usr/local/m6801-tools
```

2.4 Manual pages

Each tool has a manual page in `doc/man`. `make install` installs them (see above), with `nm01` and `size01` as links to the `objdump01` page. To install only the manual pages, use:

```
$ sudo make install-man
$ man as01
```

You can also read one without installing it, using `man -l doc/man/as01.1`.

2.5 Building the manual

This manual is built from `doc/manual.md`, with the command reference generated from the manual pages and the instruction table generated by running the assembler:

```
$ make            # build the tools first
$ cd doc
$ make
```

Chapter 3

Getting Started

This chapter builds a small program for the Tandy MC-10, which has an MC6803 processor. The program prints a message on the screen using the character output routine in the MC-10 ROM, whose address is stored at `$FFDE`.

Save this as `hello.s`:

```
; hello.s - print a message on the MC-10 screen
        LOC     0
        ENTRY   start

start   ldx     #msg
loop    ldaa    ,x              ; next character
        beq     done            ; zero byte ends the string
        inx
        pshx
        ldx     $ffde           ; ROM character output routine
        jsr     ,x
        pulx
        bra     loop
done    rts

msg     fcc     "HELLO, WORLD!"
        fcb     $0d,0
```

Labels start in column 1; instructions are indented. `ENTRY start` marks `start` as the place where the program begins.

Assemble the source into an object module:

```
$ as01 -o hello.o hello.s
```

Link it. MC-10 machine language programs are commonly loaded at `$434C`, just above the BASIC program area. The linker's `-T` option sets the start address. Like all addresses given to the tools, it is read like a C number, so write hexadecimal with `0x`:

```
$ ld01 -T0x434c -o hello.aout hello.o
```

Convert the result into a cassette audio file. `objcopy01` finds the load address in the executable, and `-n` sets the name that the MC-10 shows when it loads the file:

```
$ objcopy01 -O wav -n HELLO hello.aout hello.wav
```

Play `hello.wav` into the MC-10's cassette input (or load it into an emulator), type `CLOADM`, and then `EXEC` to run the program.

To see what the assembler produced, ask for a listing with `-l`:

```
$ as01 -o hello.o -l hello.lst hello.s
```

```
00005 0000   CE   0012> start   ldx #msg
00006 0003   A6     00  loop    ldaa    ,x      ; next character
00007 0005   27     0A      beq done        ; zero byte ends the string
...
00016 0012           48  msg fcc "HELLO, WORLD!"
```

The columns are the line number, the address, the opcode and the operand bytes. A `>` after the operand means the value is relocatable and will be adjusted by the linker. Here `msg` is at offset `$0012` from the start of the module, and becomes `$435E` once linked at `$434C`. `fcc` and `fcb` lines show only the first byte.

Chapter 4

The Assembler

4.1 Invoking as01

```
as01 [-gkuvw] [-b binfile] [-l [listfile]] [-m [listfile]]
     [-n name] [-o objfile] [-s symfile] [-t n] [srcfile]
```

Option	Meaning
`-o objfile`	Write a relocatable object file
`-b binfile`	Write an absolute binary image (see Section 4.10)
`-l [listfile]`	Write a listing to *listfile*, or to standard output
`-m [listfile]`	List macro expansions in full (see Section 4.9)
`-s symfile`	Write a symbol table file
`-n name`	Module name to store in the object file (default: from the source file name)
`-g`	Put only global symbols in the object file
`-u`	Treat undefined symbols as imported
`-w`	Turn warnings off (they are on by default; `-w-` turns them on)
`-k`	Keep output files even if there are errors
`-t n`	Put `.TEXT` sections in location counter n+3
`-v`	Print the version and exit

Each option must be a separate argument (write `-g -u`, not `-gu`). Only one source file may be given; if there is none, the source is read from standard input. If there are errors, the object and binary files are deleted (unless `-k` is given) and `as01` exits with status 1.

4.2 Source lines

Each line has up to four fields:

```
[label[:]]   [operation   [operands]]   [; comment]
```

- A **label** without a colon must start in column 1. A label followed by a colon may be indented.
- The **operation** is an instruction mnemonic, a directive or a macro name. It must be preceded by white space unless it follows a label.
- **Operands** are separated by commas.
- A **comment** starts with `;` or `!`. A line whose first character is `*`, `;` or `!` is a comment line. Blank lines are allowed.

A line consisting only of a label defines the label at the current address. If the assembler finds a word it doesn't know where an operation should be, it reports `opcode expected`.

```
* A comment line
table   fcb     1,2,3           ; label in column 1
        ldaa    table           ; instruction, indented
next:   inca                    ; label with colon
here                            ; label alone
```

4.3 Symbols

Symbols are made of letters, digits, `_` and `.`, and must not start with a digit. They are case sensitive: `Loop` and `loop` are different.

Mnemonics, directives and register names are recognized in all upper case or all lower case (`LDAA` or `ldaa`, but not `LdAa`). They cannot be used as labels. The register names are `A`, `B`, `D` and `X`.

4.4 Constants

Form	Meaning
`123`	Decimal
`$7F` or `0x7F`	Hexadecimal
`0177`	Octal (any number with a leading zero)
`%1010`	Binary
`'A` or `'A'`	Character code

4.5 Expressions

From lowest to highest precedence, the expression operators are:

Operators	Meaning
`= < >`	Comparison: true is -1 (`$FFFF`), false is 0
`+ - |`	Addition, subtraction, bitwise OR
`* / & << >>`	Multiplication, division, bitwise AND, shifts
`- + ~`	Unary minus, unary plus, bitwise complement

Use parentheses to group subexpressions. A `*` where an operand is expected stands for the current location counter, so `bra *` is an infinite loop.

A symbol is *absolute* if it has a fixed value (a constant, or a label after `ORG` in a non-relocatable program) or *relocatable* if its value depends on where the linker puts the module. You can add a constant to a relocatable value or subtract two relocatable values from the same section, but other combinations, such as adding two relocatable values, are errors.

4.6 Addressing modes

Mode	Example	Notes
Inherent	`inx`	No operand
Immediate	`ldaa #$10`	`#` prefix; 16-bit for `LDD`, `LDX`, `LDS`, `CPX`, `ADDD`, `SUBD`
Direct	`ldaa $80`	Address in `$00`–`$FF`, one byte
Extended	`ldaa $1234`	Any 16-bit address
Indexed	`ldaa 5,x`	Unsigned 8-bit offset (0–255) from X; `,x` means `0,x`
Relative	`bne loop`	Branches; target within -128 to +127 bytes

The assembler chooses between direct and extended addressing on its own. It uses direct addressing when the address is in `$00`–`$FF`, even for a symbol defined later in the source: it makes at least three passes, and more while instructions are still changing size. Relocatable addresses use extended addressing, unless their location counter is declared with `DPAGE` (see Page-zero variables). You can force the choice by writing `<` (direct) or `>` (extended) before the operand:

```
        ldaa    <port           ; direct, even if port is defined later
        ldaa    >$0080          ; extended, even though $80 fits in a byte
```

A forced direct address must fit in a byte. The assembler reports `address out of bounds` for an absolute address, or one in the module, that doesn't, and the linker checks imported and relocated addresses.

The MC6801 has no direct mode for the read-modify-write instructions (`NEG`, `COM`, `LSR`, `ROR`, `ASR`, `ASL`/`LSL`, `ROL`, `DEC`, `INC`, `TST`, `CLR`) or for `JMP`. These always use extended addressing for non-indexed operands, and `<` is an error with them.

The MC6801 has 128 bytes of internal RAM at `$0080`–`$00FF` and its I/O registers at `$0000`–`$001F`. Give them names with `EQU`, and the assembler uses direct addressing for them:

```
PORT1   equ     $02
RAM     equ     $80

        ldaa    PORT1           ; direct: 96 02
        clr     RAM             ; extended: 7F 0080 (no direct CLR)
```

4.7 Instruction set

`as01` accepts the full MC6801/MC6803 instruction set, using the mnemonics from the Motorola data sheet. Appendix A lists every instruction with its opcodes. Some instructions have two names:

Name	Same as
`CMPX`	`CPX`
`LSL`, `LSLA`, `LSLB`, `LSLD`	`ASL`, `ASLA`, `ASLB`, `ASLD`
`BHS`	`BCC`
`BLO`	`BCS`

4.8 Directives

Directives (also called pseudo-ops) control the assembler. Many have several names, for compatibility with other assemblers; they are separated by `/` below.

4.8.1 Defining data

Directive	Meaning
`FCB`/`.BYTE`/`DB`/`.DATA1` *expr*,...	Bytes: numbers, or addresses of page-zero symbols (`DPAGE`)
`FDB`/`.WORD`/`DW`/`.SHORT`/`.DATA2` *expr*,...	16-bit words, high byte first
`FCC`/`.ASCII` *string*	Characters of a string
`.ASCIZ` *string*	Characters of a string, then a zero byte
`RMB`/`.BLKB`/`.SPACE` *count*[,*fill*]	Reserve *count* bytes; filled with *fill* if given
`.BLKW`/`.ZEROW` *count*	Reserve *count* words

A string starts with a delimiter character and ends at the next occurrence of it, so `"text"`, `/text/` and `'text'` are all valid. Two delimiters in a row stand for one delimiter character. These C escape sequences are recognized: `\n`, `\r`, `\t`, `\a`, `\b`, `\f`, `\v`, `\\`, `\'`, `\"`, `\?` and octal `\0` to `\377`.

```
msg     fcc     "Hello"
        fcb     $0d,0
name    .asciz  /MC-10/
table   fdb     start,stop,$1000
buffer  rmb     32              ; 32 bytes, not initialized
spaces  rmb     8,$20           ; 8 bytes containing $20
```

4.8.2 Defining symbols

Directive	Meaning
label `EQU` *expr*	Define a constant
label `=` *expr*	Same as `EQU`
label `SET` *expr*	Define a symbol that can be redefined later
`EXPORT/PUBLIC/.DEFINE` *sym,...*	Make symbols visible to other modules
`ENTRY` *sym*	Export *sym* and mark it as the program entry point
`IMPORT/EXTERN/.EXTERN/EXTRN` *sym,...*	Use symbols from other modules
`.GLOBL/.GLOBAL` *sym,...*	Export if defined here, otherwise import
label `COMM` *size*	Common block of *size* bytes, shared between modules
label `LCOMM` *size*	Common block local to this module
`.COMM/.LCOMM` *name,size*	Unix-style forms of the above

4.8.3 Location and sections

Directive	Meaning
`LOC` *n*	Switch to location counter *n* (0–15)
`.TEXT`	Switch to location counter 0
`.DATA`, `.ROM`	Switch to location counter 3
`.BSS`	Switch to location counter 4, for uninitialized data (Section 5.2.1)
`.SECT` *name*	Switch to section `.text`, `.data` or `.bss`
`DPAGE` *n*	Location counter *n* holds page-zero data (see Page-zero variables)
`ORG` *expr*	Set the location counter to *expr* (only with `-b`)
`BLOCK` *expr* ... `ENDB`	Assemble the enclosed lines at *expr*, then return to the previous location (only with `-b`)
`.ALIGN` *n*	Advance to a multiple of *n*
`.EVEN`	Advance to an even address

Section 5.2 explains how the location counters are mapped into memory.

4.8.4 Conditional assembly

Directive	Meaning
`IF` *expr*	Assemble the following lines if *expr* is non-zero
`IFC` (*str1*,*str2*)	Assemble the following lines if the strings are identical
`ELSEIF` *expr*, `ELSEIFC` (*str1*,*str2*)	Alternative conditions

Directive	Meaning
`ELSE`	Assemble if no earlier condition was true
`ENDIF`	End of the conditional block

Conditionals may be nested. Conditions should only use symbols defined earlier in the source.

```
DEBUG   equ     1

        IF      DEBUG
        jsr     dump_registers
        ELSE
        nop
        ENDIF
```

4.8.5 Macros

A macro is defined with `MACRO` *name* and ends with `MEND`. Inside the body, `?1`, `?2`, ... are replaced by the arguments of the call. The arguments are separated by commas.

```
        MACRO   store2          ; store A and B at two addresses
        staa    ?1
        stab    ?2
        MEND

        store2  $1000,$1002
```

A macro must be defined before it is used. Within a macro, `IFC` can test the arguments, for example `IFC (?1,)` to check whether the first argument is empty.

4.8.6 Other directives

Directive	Meaning
`INCLUDE`/`GET` *file*	Assemble the contents of *file* at this point
`END`	End of the source; anything after it is ignored
`FAIL`	Report an error (useful in conditionals)
`.LIST`, `.MACLIST` [*n*]	Turn listing or macro listing on and off (only when `-l` or `-m` is used)
`.WARN` [*n*]	Turn warnings off (`.WARN`) and on again (`.WARN 1`)

4.9 Listings and symbol files

`-l` writes a listing with the source, the address and the generated bytes, followed by a symbol table and the number of errors and warnings (see the example in Chapter 3). The instructions generated by macro calls are listed after the call, marked with `+a`.

`-m` lists every line of each macro expansion, including directives and lines skipped by conditionals. On its own it lists only the macro expansions; use `-l -m listfile` to get the full listing as well.

In the listing's symbol table, each symbol is followed by its location counter number, its value, `A` (absolute) or `R` (relocatable), and `E` (exported), `I` (imported), `N` (entry point) or `C` (common).

`-s` writes a symbol file with one line per symbol: the counter, the value, five flag columns (`E` exported, `N` entry, `I` imported, `R` or `A`, `C` common) and the name:

```
0 0000 --IR- ext
0 000C EN-R- here
```

4.10 Absolute programs

For a simple program you can skip the linker entirely. Use `ORG` to set the address and `-b` to write the binary image:

```
        org     $e000
reset   lds     #$00ff
        ...
        org     $fffe
        fdb     reset           ; reset vector
```

```
$ as01 -b rom.bin rom.s
```

The image starts at the lowest address assembled and ends at the highest, so this example produces an 8 KB file covering `$E000` to `$FFFF`. Gaps between `ORG` areas are filled with zeros. `IMPORT` and other external references are not allowed with `-b`.

`ORG` and `BLOCK` are only for absolute programs. In a module for the linker, they are an error, since the linker decides where the module goes; place code at fixed addresses with a linker script instead (chapter 9).

4.11 Error messages

Errors are shown in the listing under the line that caused them, with a `^` pointing at the position. If no listing file was requested, the offending lines and the totals are printed on standard output. Some common messages:

Message	Usual cause
`opcode expected`	Misspelled or unknown instruction, or a label that isn't in column 1
`undefined label`	A symbol is used but never defined or imported
`redefined label`	The same label is defined twice
`illegal address mode`	The instruction has no such addressing mode, such as `staa #1` or `clr <$80`
`address out of bounds`	Branch target too far away, indexed offset outside 0–255, or a `<` address that doesn't fit in a byte
`data out of bounds`	Value too large for a byte
`junk after operands`	Extra text after the operands; comments must start with `;`
`register used as identifier`	`A`, `B`, `D` or `X` used as a label
`absolute expression required`	A relocatable value where a constant is needed
`no imports with binary file output`	`IMPORT` used together with `-b`
`ORG or BLOCK in relocatable module (needs -b)`	`ORG` or `BLOCK` without `-b`; use a linker script to place code
`short branch would do`	(warning) a `JSR`/`JMP` target is close enough for a `BSR`/`BRA`

Chapter 5

Modules and Linking

5.1 Exporting and importing symbols

Each source file is assembled into an object module. By default, its symbols are private. To share them:

- In the module that defines a symbol, name it in an `EXPORT` (or `ENTRY`) directive.
- In each module that uses it, name it in an `IMPORT` directive.

```
; screen.s
        LOC     0

clrscn  ldx     #$4000
        ...
        rts
        EXPORT  clrscn
```

```
; main.s
        LOC     0
        IMPORT  clrscn
        ENTRY   start
start   jsr     clrscn
        rts
```

The linker reports `undefined symbol` for any import that no module exports.

Exactly one module should contain an `ENTRY` directive. The linker uses the first entry symbol it finds as the program's starting address, and fails with `no start symbol` if there is none.

5.2 Location counters and segments

The assembler keeps 16 independent location counters, numbered 0 to 15, selected with `LOC` or with `.TEXT`, `.DATA` and `.BSS`. Code assembled under different counters is kept apart, even if the source switches between them, and each module's counters start at 0.

The linker gathers all modules' counters into segments:

Counters	Segment
0 and 1	Text, placed at the `-T` address

Counters	Segment
2, 3 and 5 to 15	Data, placed after the text, or at the `-D` address
4, and all common blocks	Uninitialized data, after the data, not in the file

`.TEXT` selects counter 0, `.DATA` and `.ROM` select counter 3, and `.BSS` selects counter 4. Each module's part of a segment follows the one before it, with no gap. If counter 2 would cross a 256-byte page boundary, the linker moves it to the start of the next page.

Most MC6801 programs need only counter 0. Start each module with `LOC 0` (or `.TEXT`) to make this explicit.

This is the default layout. A linker script can place each module's counters anywhere (Chapter 9).

5.2.1 Uninitialized data

Space reserved with `RMB` under `.BSS`, and common blocks (`COMM`), are *uninitialized data*: the default layout gives them addresses after the data, but they aren't stored in the executable, so they take no space in the file, on cassette or in ROM. The a.out header records their size (the `bss` column of `size01`).

Since they aren't loaded, they hold whatever was in memory before the program started. A program that needs them to be zero must clear them in its start-up code. The linker provides `__bss_start` and `__bss_end`, the start and end of the `.bss` section, for this:

```
        IMPORT  __bss_start, __bss_end
        ldx     #__bss_start    ; clear .BSS and the common blocks
clear   cpx     #__bss_end
        beq     cleared
        clr     0,x
        inx
        bra     clear
cleared
```

Only reserved space can go under `.BSS`. Initialized data there, such as `FCB`, would be lost, so the linker reports it:

```
ld01: main.o: LOC4 has initialized data, but .bss is NOLOAD
```

5.3 Page-zero variables

The MC6801's page zero (`$0000`–`$00FF`) holds its I/O registers and its 128 bytes of internal RAM, at `$0080`–`$00FF`. Instructions reach it with *direct* addressing, a one-byte address, which saves a byte and a cycle over extended addressing. The assembler can only use direct addressing for an address

it knows, though, and the address of a variable that the linker places isn't known until the program is linked.

`DPAGE` *n* tells the assembler that location counter *n* holds page-zero data. Put the variables in that counter, and declare it with `DPAGE`, near the top of every module that uses them (it applies from where it appears):

```
; zpvars.s - variables in page zero, in location counter 5
        DPAGE   5
        LOC     5
        EXPORT  count, flags
count   rmb     1
flags   rmb     1

; zpmain.s - uses them
        DPAGE   5
        LOC     5
        IMPORT  count, flags    ; imported while counter 5 is current
        LOC     0
start   ldaa    count           ; 96 xx: direct
        staa    flags           ; 97 xx: direct
        ldab    >count          ; F6 xxxx: extended, forced with >
        inc     count           ; 7C xxxx: INC has no direct mode
        ldx     #count          ; CE xxxx: a 16-bit immediate
        ldab    #flags          ; C6 xx: the address as a byte
        ldaa    total           ; 96 xx: direct, though total comes later
        rts

        LOC     5
total   rmb     2
```

- **Which symbols.** A symbol is in page zero if it is defined in a `DPAGE` counter, imported while that counter is current, or declared `COMM` in it. An `IMPORT` records the current location counter, so import page-zero variables under their counter, as `zpmain.s` does. If the module that defines a variable uses a different counter, the linker warns, as in `counter is imported under LOC0 in main.o, but defined in LOC5 in vars.o`, and the importing module doesn't get direct addressing for it.
- **Direct addressing** is used wherever the instruction has a direct mode. `>` still forces extended addressing, and the read-modify-write instructions, such as `INC`, have no direct mode.
- **Addresses as bytes.** The address of a page-zero symbol also fits where a byte is expected: in 8-bit immediates, such as `ldab #flags`, and in `FCB` data.
- **Forward references.** A symbol used before it is defined, imported or declared gets direct addressing too, as `total` does: the assembler finds out in a later pass that it is in page zero.

The linker script must put the counter, and its common blocks, in page zero, for example with `.zpage (NOLOAD) : { *(LOC5) *(COMMON5) } > ZP`, where `ZP` is a `MEMORY` region at `$0080`. The linker checks every one-byte address: if the script puts one outside page zero, the link fails with an error such as

```
ld01: zpmain.o: count ($5004) is not in page zero ($00-$ff), at $e001
```

A program whose RAM is all in page zero can declare its data counter itself, as the ROM example in section 9.4 does with `DPAGE 3`. On the MC-10, the ROM and BASIC keep their own variables in the internal RAM, so a program that returns to BASIC must leave those bytes alone.

5.4 Invoking ld01

```
ld01 [-dimMPrstyZ] [-o outfile] [-S script] [-A symbol=value]
     [-T textaddr] [-D dataaddr] [-H heapsize]
     [-Ccrt] [-Llibdir] [-llib] [-Olibfile] file ...
```

Option	Meaning
`-o outfile`	Output file name (default `a.out`)
`-T addr`	Text segment address
`-D addr`	Data segment address
`-H size`	Add *size* to the end of the program for `__heap_top`
`-Cname`	Link the start-up module `crtname.o`
`-lname`	Search the library `libname.a`
`-Ldir`	Add a library directory
`-Ofile`	Link *file*, looking in the library directories
`-d`	Write a raw memory image with no header or symbols
`-s`	Leave out the symbol table
`-m`	List the modules linked
`-M`	Show the layout, then all symbols with their values
`-S script`	Link with a linker script (Chapter 9)
`-A sym=value`	Define a symbol for the linker script
`-P`	Print the default linker script and exit
`-Z`	Make input that a linker script doesn't place an error
`-r`	Combine the input objects into an archive instead of linking
`-i`	Mark the executable as having separate instruction and data spaces
`-t`	Trace modules as they are read (must come before the input files)

A module is linked only if the program uses it: starting from the module with the entry point, the linker includes every module that defines a symbol a linked module refers to. A common block (`COMM`) that another module has too is no reason to include that module: the storage comes with a linked module that has the common block, as large as the largest of any module's. The order of objects and libraries on the command line doesn't matter for finding symbols, since the linker reads them all first, but the modules are placed in that order.

Addresses and sizes are read like C numbers: hexadecimal with a `0x` or `$` prefix, octal with a leading `0`, and decimal otherwise. So `-T0x1000`, `-T'$1000'` and `-T4096` all mean `$1000` (quote `$` from the shell). Anything else, such as `-TE000`, or a value above `$FFFF`, is an error. `objcopy01` reads its `-T` and `-D` options the same way.

5.5 Output

The linker normally writes an a.out file: a 32-byte header (Appendix B), the text segment, the data segment and a symbol table. `objcopy01` converts this into the format your target needs (Chapter 8).

With `-d`, the output is a raw memory image containing just the text segment followed by the data segment, ready to be burned into an EPROM:

```
$ ld01 -d -T0xe000 -o rom.bin start.o main.o
```

A linker script can also ask for a memory image, S-records or Intel hex with `OUTPUT_FORMAT` (Section 9.6).

5.6 Start-up modules

A *start-up module* is code that must run first: setting up the stack, clearing memory, then calling the main program. By convention it is called `crtname.o` (from "C run-time") and linked with `-Cname`:

```
; crtstart.s
        LOC     0
        IMPORT  main
        ENTRY   __start
__start lds     #$00ff
        jsr     main
        rts

$ as01 -o crtstart.o crtstart.s
$ ld01 -Cstart -T0x434c -o prog.aout main.o
```

5.7 Libraries

`-lname` links the needed members of the archive `libname.a`. An archive member is linked only if it defines a symbol that something else being linked uses. See Chapter 6 for how to build libraries.

Files named with `-C`, `-l` and `-O` are searched for in the `-L` directories (up to five), most recent first. There is no default library directory. If they are not found, the name is tried as given, in the current directory.

5.8 Linker symbols

The linker defines these symbols, which a program can `IMPORT`:

Symbol	Value
`__btext`, `__etext`	Start and end of the text segment
`__bdata`, `__edata`	Start and end of the data segment
`__end`	End of the program, including uninitialized data
`__heap_top`	`__end` plus the `-H` value

For example, a start-up module can clear uninitialized data between `__edata` and `__end`.

`__btext`, `__etext`, `__bdata`, `__edata` and `__end` are always written to the executable's symbol table (unless `-s` is used), where `objcopy01` finds the load addresses.

5.9 Link maps

`-m` prints a line for each module: `+` if it was linked, `-` if it is an archive member that was not needed. `-M` prints the layout, each output section with its address and size and the parts of the modules in it, then every symbol with the module that defines it, its value and its type:

```
$ ld01 -lnum -Cstart -T0x434c -m -M -o main.aout main.o screen.o
one             +libnum.a(one.o)
two             +libnum.a(two.o)
three           +libnum.a(three.o)
crtstart        +crtstart.o
main            +main.o
screen          +screen.o
Section         Address  Size     Load
.text           0000434c 00000063
    0000434c 0000000f libnum.a:one.o(LOC0)
    0000435b 0000000f libnum.a:two.o(LOC0)
    0000436a 00000011 libnum.a:three.o(LOC0)
    0000437b 00000004 crtstart.o(LOC0)
    0000437f 0000000d main.o(LOC0)
    0000438c 00000023 screen.o(LOC0)
.data           000043af 00000000
.bss            000043af 00000000 (NOLOAD)

                 one               msg_one  0  0000434c  r
                 one                   one  0  00004351  R
...
```

The modules are placed in command-line order: `-lnum` comes first, so the library members are at the start. Each module's part follows the one before it, with no gap. `.bss` holds the uninitialized data (`.BSS` and common blocks), which takes no space in the file; this program has none.

Chapter 6

Libraries

A library is an archive of object modules, created and maintained with `ar01`. The usual name is `libname.a`, so that it can be linked with `-lname`.

```
$ ar01 rc libnum.a one.o two.o three.o        # create or update
$ ar01 t libnum.a                             # list members
one.o
two.o
three.o
$ ar01 d libnum.a two.o                       # delete a member
$ ar01 x libnum.a one.o                       # extract a member
```

Key	Operation
`r`	Replace or add members (creates the archive if necessary)
`q`	Append members without checking for duplicates
`d`	Delete members
`m`	Move members (to the end, or with `a`/`b` next to another member)
`t`	List members (`tv` for details)
`x`	Extract members
`p`	Print members to standard output

Modifiers: `c` (create silently), `v` (verbose), `u` (replace only newer files), `o` (keep dates on extraction), `a`/`b`/`i` *posname* (position), and `s`, which adds a `__.SYMDEF` index of the symbols each member exports. The linker doesn't need the index, since it reads every member's symbols itself, but it does no harm.

Put each routine that may be needed separately in its own module, so that the linker includes only what a program uses.

Chapter 7

Inspecting Object Files and Executables

`objdump01` shows the header, segment sizes, symbol table and a hex dump of an a.out executable:

```
$ objdump01 main.aout
OBJECTFILE 'main.aout'
HLEN 32
CPU  m6801 BIG_ENDIAN
FLAGS: A_EXEC
a_entry  = 0x0000437b
a_total  = 0x00000063
a_syms   = 0x00000140

text    data    bss     dec     hex     filename
99      0       0       99      63      main.aout

SYMBOLS
0000434c t msg_one
00004351 T one
...
TEXTSEG
: 4f 4e 45 0d 00 ce 43 4c  3c bd 43 9d 31 31 39 54  ONE...CL<.C.119T
...
```

`nm01` (or `objdump01 -n`) prints only the symbols, and `size01` (or `objdump01 -s`) only the sizes:

```
$ size01 main.aout
text    data    bss     dec     hex     filename
99      0       0       99      63      main.aout
$ nm01 main.aout
0000434c t msg_one
00004351 T one
```

```
0000437b T __start
00004000 a SCREEN
...
```

In the symbol list, `T/t` are text symbols, `D/d` data symbols, `C/b` bss or common symbols and `a` absolute symbols. Upper case means global, lower case local, and `U` marks a symbol that is used but not defined. Names of any length are kept in executables (`ld01 -y-` writes the older symbol table, which cuts them to 8 characters).

The same tools work on object files made by `as01`, and on libraries made by `ar01`, which shows each module in turn. For an object file, `objdump01` shows the module's symbols, its size, and its code with the relocation records that `ld01` uses to fill in addresses:

```
$ objdump01 main.o
OBJECTFILE 'main.o'
MODULE  'main'
...
SYMS 6
SYM 0    00000000 ..E.T main
SYM 1    00000000 ...IT three
...
BYTECODE
SEG 0
: bd                                                    .
RELSZ 2
DW clrscn
...
$ nm01 main.o
00000000 T main
         U three
         U one
...
```

Here `DW clrscn` means that the two bytes after the `jsr` opcode (`bd`) are filled in with the address of `clrscn`. In object files, location counters 0 and 1 count as text, counter 4 (`.BSS`) as bss, and the others as data, as they are linked by `ld01`.

7.1 Disassembly

`objdump01 -d` also disassembles the code. In an object file, each operand that the linker fills in is shown by name: the imported symbol, or the label nearest before the address in the module:

```
$ objdump01 -d one.o
...
DISASSEMBLY
SEG 0
```

```
0000  4f        msg_one  clra
0001  4e                 fcb    $4e
0002  45                 fcb    $45
0003  0d                 sec
0004  00                 fcb    $00
0005  ce 00 00  one      ldx    #msg_one
0008  3c                 pshx
0009  bd 00 00           jsr    prstring
000c  31                 ins
000d  31                 ins
000e  39                 rts
```

The disassembler can't tell data from code, so the string `ONE` at `msg_one` comes out partly as instructions (`4f` is `clra`), and partly as `fcb` where its bytes aren't opcodes. In an executable, the code is at its linked addresses, with labels from the symbol table, and calls and branches are shown by name:

```
$ objdump01 -d main.aout
...
437b  bd 43 7f  __start  jsr    main
437e  39                 rts
...
438c  ce 40 00  clrscn   ldx    #$4000
438f  86 60              ldaa   #$60
4391  c6 60              ldab   #$60
4393  ed 00     clrsc.1  std    $00,x
4395  08                 inx
4396  08                 inx
4397  8c 42 00           cpx    #$4200
439a  2d f7              blt    clrsc.1
439c  39                 rts
```

A direct address that the linker fills in is shown with `<`, such as `ldaa <count` for a page-zero variable, and an extended address in page zero with `>`, as you would write them for `as01`.

Chapter 8

Converting Executables

`objcopy01` turns an a.out file into a format that a loader, EPROM programmer, emulator or cassette interface can use:

```
objcopy01 [-O format] [-T textaddr] [-D dataaddr] [-n name] [-p]
          infile outfile
```

Format	Output
`binary`	Raw image: text, then data (`-p` puts each at its address)
`srec`	Motorola S-records (S0 name, S1 data, S5 count, S9 entry point)
`ihex`	Intel hex, with the entry point in the end record
`wav`	Cassette audio: 8-bit mono WAV at 43200 Hz, Color BASIC/MC-10 format
`cas`, `c10`	The same cassette data as a byte stream, for emulators (`c10` matches the name used by MC-10 emulators)

A program's data needn't follow its text: `ld01 -D`, or a linker script, can put it elsewhere, above the text or below it. `srec` and `ihex` write each part at its address. `binary`, `wav` and `cas` hold one block, so they need `-p`, which writes one image from the lower address to the end of the higher, with zeros between (without it, `wav` and `cas` refuse such a program).

The load addresses come from the `__btext` and `__bdata` symbols that the linker writes to the executable, so you don't normally need to give them. Use `-T` and `-D` to override them, or when the executable was linked with `ld01 -s` and has no symbol table. Addresses are written the same way as for `ld01`: `0x434c` or `'$434c'` for hexadecimal, plain digits for decimal.

The entry point is taken from the executable header. `-n` sets the name stored in S-record headers and cassette files (up to 8 characters for cassettes); without it, the output file name in upper case is used.

```
$ objcopy01 -O srec -n ROM rom.aout rom.s19
$ objcopy01 -O wav -n GAME game.aout game.wav
$ objcopy01 -O srec -T 0xe000 -n ROM stripped.aout rom.s19
```

The cassette formats require the data segment to follow the text segment directly (no `-D`).

`ld01` can also write S-records, Intel hex and memory images itself, with `OUTPUT_FORMAT` in a linker script (Section 9.6). That works for programs that an a.out file can't hold, such as a ROM program with its vector table at `$FFF0`.

Chapter 9

Linker Scripts

The default layout, code then data, suits a program that's loaded into RAM in one piece, like the MC-10 programs in this manual. Other programs need something else: a ROM program has its code in ROM, its variables in RAM and its interrupt vectors at the top of memory. A *linker script*, given with `-S`, says where everything goes.

This chapter introduces linker scripts by example. The LINKER SCRIPTS section of the `ld01` manual page (Appendix C) is the complete reference.

9.1 Input sections and output sections

A script works with two kinds of section:

- An **input section** is one location counter of one module. Each module has `LOC0` to `LOC15`, its code and data in each counter, and `COMMON0` to `COMMON15`, its common blocks. `.text` is another name for `LOC0`, `.data` for `LOC3`, and `COMMON` means all the common blocks.
- An **output section** is a named part of the program, such as `.text`, with an address. The script lists the input sections that go into it.

An input section is named by a pattern: the file, then the section names in parentheses. `screen.o(LOC0)` is counter 0 of `screen.o`, `*(LOC0 LOC1)` is counters 0 and 1 of every module, and `libnum.a:one.o(LOC*)` is every counter of one library member. The wildcards `*` and `?` work in both parts.

Each input section goes to the first pattern in the script that matches it, so specific patterns come before general ones. A pattern adds its input sections in module order (the order of the command line), and each module's in counter order.

A pattern can leave files out with `EXCLUDE_FILE`, which names the files (patterns too) before the pattern they apply to. The vector table can then go anywhere in the script:

```
.text :
{
    crt0.o(LOC0)
```

```
        EXCLUDE_FILE(vectors.o) *(LOC0 LOC1)
    } > ROM

    .vectors :
    {
        KEEP(vectors.o(LOC0))
    } > VEC
```

9.2 Starting from the default layout

The default layout is itself a script, built into `ld01`. `-P` prints it, with the values of `-T`, `-D` and `-H` filled in:

```
$ ld01 -P -T0x434c
/* Default linker script for ld01.
   Values from the command line: -T 0x434C, no -D, no -H. */

OUTPUT_FORMAT(aout)

SECTIONS
{
    .text 0x434C :
    {
        __btext = .;
        PROVIDE(__segoff = 0);
        PROVIDE(__seg0DL = .);
        *(LOC0)
...
```

`.text 0x434C` is an output section at `$434C`, and `*(LOC0)` adds counter 0 of every module to it. `.` is the current address, so `__btext = .` defines `__btext` as the start of the section. Linking with this script (`-S`) gives exactly the same output as linking without one, so it's a safe starting point for your own.

9.3 A program for the MC-10

This script links the demo program from Chapter 10 for the MC-10. It puts the start-up code first, then the rest of the code, then the data:

```
/* mc10.ld - a program for the Tandy MC-10, loaded from cassette at
   $434C: the start-up code first, then the rest of the code, then the
   data */
ENTRY(__start)

SECTIONS
```

```
{
    .text 0x434C :
    {
        __btext = .;
        crtstart.o(LOC0)
        *(LOC0 LOC1)
        *(COMMON0 COMMON1)
        __etext = .;
    }

    .data :
    {
        __bdata = .;
        *(LOC*)
        *(COMMON)
    }

    __stack_top = 0x4FFF;
}
```

```
$ ld01 -S mc10.ld -lnum -o main.aout crtstart.o main.o screen.o
$ objcopy01 -O wav -n MAIN main.aout main.wav
```

- `crtstart.o(LOC0)` comes before `*(LOC0 LOC1)`, so the start-up code goes first. `crtstart.o` is named on the command line here, but `-Cstart` would work too.
- `*(LOC*)` in `.data` also takes counter 4, `.BSS`, so any uninitialized data is stored in the program as zeros, where the default layout leaves it out (see section 5.2).
- `__btext`, `__etext` and `__bdata` are defined so that `objcopy01` can find the load address. The default script defines them; your own script should too if you'll convert its output with `objcopy01`.
- `__stack_top` is a symbol that a start-up module can import, so the stack's address is set in one place.

`-M` shows the result:

```
Section         Address  Size     Load
.text           0000434c 00000063
    0000434c 00000004 crtstart.o(LOC0)
    00004350 0000000f libnum.a:one.o(LOC0)
    0000435f 0000000f libnum.a:two.o(LOC0)
    0000436e 00000011 libnum.a:three.o(LOC0)
    0000437f 0000000d main.o(LOC0)
    0000438c 00000023 screen.o(LOC0)
.data           000043af 00000000
```

Each line under a section is one input section: its address, its size, and the module and counter it came from.

9.4 A program in ROM

This example is a complete program for a bare MC6801 system with an 8 KB EPROM at `$E000` and the MC6801's 128 bytes of internal RAM at `$0080`. The program shows a changing pattern on port 1:

```
; prog.s - a tiny program: shows a changing pattern on port 1
        DPAGE   3
PORT1   equ     $02
        LOC     3
pattern fcb     $55
count   COMM    1
        LOC     0
        EXPORT  main
main    ldaa    pattern
loop    staa    PORT1
        coma
        inc     count
        bra     loop
```

`pattern` is initialized data, and `count` is an uninitialized variable. Both must be in RAM, but RAM is empty when the system starts, so the start-up code copies the initialized data from a copy in ROM. All of the program's RAM is the MC6801's internal RAM, in page zero, so both modules declare their data counter with `DPAGE 3`, and define their variables before the code that uses them. Their references to `src`, `dst` and `pattern` then use direct addressing:

```
; crt0.s - start-up code for a ROM program: sets up the stack, copies
; the initialized data from ROM to RAM, and calls main
        DPAGE   3               ; this program's data is all in page zero
        LOC     3
src     COMM    2               ; copy pointers
dst     COMM    2
        LOC     0
        ENTRY   reset
        EXPORT  ignore
        IMPORT  __stack_top, __data_load, __data_start, __data_end, main
reset   lds     #__stack_top
        ldx     #__data_start   ; any data to copy?
        cpx     #__data_end
        beq     run
        ldx     #__data_load    ; copy it from ROM to RAM
        stx     src
        ldx     #__data_start
        stx     dst
copy    ldx     src
        ldaa    0,x
        inx
```

```
        stx     src
        ldx     dst
        staa    0,x
        inx
        stx     dst
        cpx     #__data_end
        bne     copy
run     jsr     main
        bra     *

ignore  rti                     ; for interrupts that aren't used
```

The MC6801 finds the address of the reset routine, and of each interrupt routine, in a table at `$FFF0` to `$FFFF`:

```
; vectors.s - the MC6801 interrupt vectors, for $FFF0 to $FFFF
        LOC     0
        IMPORT  reset, ignore
        fdb     ignore          ; SCI
        fdb     ignore          ; timer overflow
        fdb     ignore          ; output compare
        fdb     ignore          ; input capture
        fdb     ignore          ; IRQ1
        fdb     ignore          ; SWI
        fdb     ignore          ; NMI
        fdb     reset           ; reset
```

The linker script puts each part in the right place:

```
/* rom.ld - a program in an 8 KB EPROM at $E000, with its variables in
   the MC6801's internal RAM */
ENTRY(reset)
OUTPUT_FORMAT(srec)
FILL(0xFF)

MEMORY
{
    ROM (rx) : ORIGIN = 0xE000, LENGTH = 0x1FF0
    VEC (r)  : ORIGIN = 0xFFF0, LENGTH = 16
    RAM (rw) : ORIGIN = 0x0080, LENGTH = 128
}

SECTIONS
{
    .vectors :
    {
        KEEP(vectors.o(LOC0))
```

```
    } > VEC

    .text :
    {
        crt0.o(LOC0)
        *(LOC0 LOC1)
    } > ROM

    .data :
    {
        *(LOC2 LOC3 LOC5 LOC6 LOC7 LOC8 LOC9 LOC1?)
    } > RAM AT> ROM

    .bss (NOLOAD) :
    {
        *(.bss)
        *(COMMON)
    } > RAM

    __stack_top = ORIGIN(RAM) + LENGTH(RAM) - 1;

    ASSERT(SIZEOF(.vectors) == 16, "the vector table must be 16 bytes")
}

$ as01 -o crt0.o crt0.s
$ as01 -o vectors.o vectors.s
$ as01 -o prog.o prog.s
$ ld01 -S rom.ld -o rom.s19 crt0.o vectors.o prog.o
```

- **MEMORY** describes the system's memory as three regions. A section placed in a region with `> ROM` starts at the region's next free address, and the linker reports an error if it doesn't fit.
- **.vectors** uses KEEP, because no module refers to `vectors.o`. Without it, the linker would leave the module out. `.vectors` comes first in the script because `*(LOC0 LOC1)` in `.text` would otherwise match `vectors.o` too. (Or `.text` could leave it out with `EXCLUDE_FILE(vectors.o) *(LOC0 LOC1)`.)
- **.data** runs in RAM (`> RAM`) but is stored in ROM after the code (`AT> ROM`). The linker provides `__data_load`, `__data_start` and `__data_end`, the ROM and RAM addresses `crt0.s` copies between, because `crt0.s` imports them. Every output section gets such symbols: `__name_start`, `__name_end` and `__name_load`.
- **.bss** holds .BSS (`*(.bss)`, counter 4) and the common blocks. `NOLOAD` means it isn't written to the output: the variables only need addresses in RAM. A `NOLOAD` section can only hold common blocks and space reserved with `RMB`: initialized data there would be lost, so the linker reports it as an error.
- **__stack_top** is the last byte of RAM, from `ORIGIN` and `LENGTH`.
- **ASSERT** checks that the vector table is exactly 16 bytes.
- **DPAGE 3** works because the script puts all of counter 3, and its common blocks, in `RAM`, which is in page zero. It makes the code 7 bytes smaller (`$38` bytes instead of `$3F`), and the copy loop 4 cycles faster for each byte. If the script put `RAM` anywhere else, the link would fail.

- `FILL(0xFF)` fills gaps with `$FF`, like an erased EPROM.
- `OUTPUT_FORMAT(srec)` writes Motorola S-records. An a.out file can't hold this program: its parts are at `$E000` and `$FFF0`, and `.data` is stored somewhere other than where it runs.

The layout:

```
Memory region   Origin   Length   Used
ROM             0000e000 00001ff0 00000039
VEC             0000fff0 00000010 00000010
RAM             00000080 00000080 00000006

Section         Address  Size     Load
.vectors        0000fff0 00000010
    0000fff0 00000010 vectors.o(LOC0)
.text           0000e000 00000038
    0000e000 0000002e crt0.o(LOC0)
    0000e02e 0000000a prog.o(LOC0)
.data           00000080 00000001 0000e038
    00000080 00000001 prog.o(LOC3)
.bss            00000081 00000005 (NOLOAD)
    00000081 00000004 crt0.o(COMMON3)
    00000085 00000001 prog.o(COMMON3)
```

The **Load** column shows that `.data` runs at `$0080` but is stored at `$E038`, right after the code. The S-records have one run for each part written to the output, with the entry point, `reset`, in the last record:

```
S00A0000726F6D2E7331399C
S113E0008E00FFCE00808C0081271DCEE038DF8398
S113E010CE0080DF81DE83A60008DF83DE81A700D7
S113E02008DF818C008126EDBDE02E20FE3B96802A
S10BE0309702437C008520F8EF
S104E038558E
S113FFF0E02DE02DE02DE02DE02DE02DE02DE000C2
S5030006F6
S903E0001C
```

9.5 Defining symbols

A script can define symbols that modules import:

- `name = expression;` always defines `name`.
- `PROVIDE(name = expression);` defines it only if a module imports it and no module defines it.
- `HIDDEN(name = expression);` defines a symbol for the script's own use, which modules can't see.

- `-A name=value` on the command line defines a symbol before the script runs. It takes the place of a `PROVIDE` for the same symbol, which makes `PROVIDE` a way to give a default. With this in a script, `__stack_top` is `$4FFF` unless the command line says otherwise:

```
PROVIDE(__stack_top = 0x4FFF);
```

```
$ ld01 -S prog.ld -A __stack_top=0x3FFF -o prog.aout start.o prog.o
```

Expressions use C operators, `.` for the current address, and functions such as `ADDR(.text)`, `SIZEOF(.data)`, `ALIGN(16)` and `ORIGIN(RAM)`. Symbols defined by a script are absolute, and satisfy an `IMPORT` from any location counter.

9.6 Output formats

`OUTPUT_FORMAT` chooses the kind of output file:

Format	Output	Use it for
`aout`	a.out executable (the default)	Programs to convert with `objcopy01`, for example to cassette
`binary`	Memory image, from the lowest load address to the highest	EPROM programmers and emulators that take raw images
`srec`	Motorola S-records	EPROM programmers and loaders; programs in several parts
`ihex`	Intel hex	The same, for tools that take Intel hex

An a.out file has just two parts, text and data, so it can't hold a program spread over separate areas of memory, or a section stored away from where it runs. Use one of the other formats for those.

9.7 Orphans

If the script doesn't place some input, for example because it only lists `LOC3` and a module also uses `LOC5`, the linker adds it to the last output section of the same kind that takes input from all files (a `*` pattern), and warns:

```
ld01: warning: segs.o(LOC5) isn't placed by the script; adding it to .data
```

`-Z` makes this an error, for builds where everything must be placed deliberately.

9.8 Things to watch for

- **Order matters.** The first matching pattern wins, so put sections that take particular modules, such as a vector table, before sections with catch-all patterns, or leave those modules out of the catch-all patterns with `EXCLUDE_FILE`. Input that a section excludes and no other pattern takes is an orphan: it goes after a section that takes all files without `EXCLUDE_FILE`, not back into that section.
- **`objcopy01` needs `__btext`.** If you convert a script's a.out output with `objcopy01`, define `__btext`, `__etext` and `__bdata` as the default script does, or give `objcopy01` the addresses with `-T` and `-D`.
- **Page zero.** Variables that a script puts in page zero (`$0000`–`$00FF`) get direct addressing only if the modules that use them declare their location counter with `DPAGE`, or write `<` before each reference (see Page-zero variables). The linker checks that every such address ends up in page zero.

Chapter 10

A Worked Example

The `demo` directory contains a small modular program for the MC-10. It shows modules, a library and a start-up module working together. With the tools installed and on your `PATH`, build it with:

```
$ cd demo
$ make
```

10.1 Modules

`screen.s` provides two subroutines, `clrscn` and `prstring`, and exports them:

```
LOC 0

SCREEN  equ     $4000

clrscn  ldx     #SCREEN
        ldaa    #$40+'
        ldab    #$40+'
clrsc.1 std     ,x
        inx
        inx
        cpx     #(SCREEN+512)
        blt     clrsc.1
        rts

        EXPORT  clrscn

prstring
        tsx
        ldx     2,x
prstlop ldaa    ,x
        beq     prstxit
        inx
```

```
        pshx
        ldx     $ffde
        jsr     ,x
        pulx
        bra     prstlop
prstxit rts

        EXPORT  prstring
```

`prstring` takes a pointer to the string on the stack. `tsx` points X at the top of the stack, and `ldx 2,x` loads the argument, which is just above the return address.

10.2 A library

`one.s`, `two.s` and `three.s` each print a word. Here is `one.s`:

```
LOC 0

        IMPORT  prstring

msg_one fcb     'O,'N,'E,$0d,$00

one
        ldx     #msg_one
        pshx
        jsr     prstring
        ins
        ins
        rts

        EXPORT  one
```

They are assembled and collected into `libnum.a`:

```
$ as01 -o one.o one.s
$ as01 -o two.o two.s
$ as01 -o three.o three.s
$ ar01 r libnum.a one.o two.o three.o
```

10.3 The start-up module and main program

`crtstart.s` contains the entry point and calls `main`:

```
LOC 0

        IMPORT  main

        ENTRY   __start

__start
        jsr     main
        rts
```

`main.s` uses the routines from `screen.o` and the library:

```
LOC 0

        IMPORT  clrscn
        IMPORT  prstring

        IMPORT  one
        IMPORT  two
        IMPORT  three

main
        jsr     clrscn

        jsr     one
        jsr     two
        jsr     three
        rts

        EXPORT  main
```

10.4 Linking and converting

```
$ ld01 -lnum -Cstart -T0x434c -o main.aout main.o screen.o
$ objcopy01 -T0x434c -O wav main.aout main.wav
```

The linker pulls in `crtstart.o` because of `-Cstart`, and `one.o`, `two.o` and `three.o` from `libnum.a` because `main.o` uses them. Load `main.wav` on the MC-10 with `CLOADM` and run it with `EXEC`.

Chapter 11

Migrating from 1.x

Version 2.0 changes where `ld01` puts things, so a program linked with the default layout ends up at different addresses than with 1.x, and usually smaller:

- **`.BSS` uses location counter 4**, not 3, and the default layout leaves it, with the common blocks, out of the executable. A program that relies on that memory being zero must now clear it at start-up (Section 5.2.1). Writing `.DATA` instead of `.BSS` gives the 1.x placement.
- **No padding.** Each module's part of a section used to be padded to a multiple of 4 bytes, and each common block rounded up to 4 bytes; neither is any more.
- **Counter 4** is `.BSS`. A program that keeps initialized data in counter 4 with the default layout gets a link error; move it to another counter. (The 1.x manual suggested counter 4 for page-zero data; this manual uses counter 5.)
- **Warnings are on.** `as01` reports warnings unless `-w` is given. Warnings don't change the exit status or the output.
- **`.bss`** in a linker script pattern means counter 4, as `.data` means counter 3.

To link with 1.x's layout, save the default script printed by `ld01 -P` from version 1.6 (with your `-T`, `-D` and `-H` options), and link with it using `-S`. It gives the same output as 1.6, except for `.BSS` data (which is in counter 4) and common blocks (which are no longer rounded up).

Chapter 12

Known Limitations

- **`objdump01 -d` can't tell data from code**, so data in a code segment, such as a string, is shown as instructions where its bytes happen to be opcodes.

Appendix A

Instruction Set

The table lists the opcode (in hexadecimal) for each instruction in each addressing mode. It was produced by assembling every instruction with `as01`.

INH = inherent, IMM = immediate, DIR = direct, IDX = indexed, EXT = extended, REL = relative.

Instruction lengths: inherent 1 byte; direct, indexed and relative 2 bytes; extended 3 bytes; immediate 2 bytes, or 3 for `ADDD`, `CPX`, `CMPX`, `LDD`, `LDS`, `LDX` and `SUBD`.

Mnemonic	INH	IMM	DIR	IDX	EXT	REL	Operation
ABA	1B						Add B to A
ABX	3A						Add B to X (unsigned)
ADCA		89	99	A9	B9		Add with carry to A
ADCB		C9	D9	E9	F9		Add with carry to B
ADDA		8B	9B	AB	BB		Add to A
ADDB		CB	DB	EB	FB		Add to B
ADDD		C3	D3	E3	F3		Add to D
ANDA		84	94	A4	B4		AND into A
ANDB		C4	D4	E4	F4		AND into B
ASL				68	78		Arithmetic shift left memory
ASLA	48						Arithmetic shift left A
ASLB	58						Arithmetic shift left B
ASLD	05						Arithmetic shift left D
ASR				67	77		Arithmetic shift right memory
ASRA	47						Arithmetic shift right A
ASRB	57						Arithmetic shift right B
BCC						24	Branch if carry clear
BCS						25	Branch if carry set
BEQ						27	Branch if equal
BGE						2C	Branch if greater or equal (signed)
BGT						2E	Branch if greater (signed)
BHI						22	Branch if higher (unsigned)
BHS						24	Branch if higher or same (= BCC)
BITA		85	95	A5	B5		Bit test A
BITB		C5	D5	E5	F5		Bit test B
BLE						2F	Branch if less or equal (signed)
BLO						25	Branch if lower (= BCS)

Mnemonic	INH	IMM	DIR	IDX	EXT	REL	Operation
BLS						23	Branch if lower or same (unsigned)
BLT						2D	Branch if less (signed)
BMI						2B	Branch if minus
BNE						26	Branch if not equal
BPL						2A	Branch if plus
BRA						20	Branch always
BRN						21	Branch never
BSR						8D	Branch to subroutine
BVC						28	Branch if overflow clear
BVS						29	Branch if overflow set
CBA	11						Compare B with A
CLC	0C						Clear carry
CLI	0E						Clear interrupt mask
CLR				6F	7F		Clear memory
CLRA	4F						Clear A
CLRB	5F						Clear B
CLV	0A						Clear overflow
CMPA		81	91	A1	B1		Compare A
CMPB		C1	D1	E1	F1		Compare B
CMPX		8C	9C	AC	BC		Compare X (= CPX)
COM				63	73		Complement memory
COMA	43						Complement A
COMB	53						Complement B
CPX		8C	9C	AC	BC		Compare X
DAA	19						Decimal adjust A
DEC				6A	7A		Decrement memory
DECA	4A						Decrement A
DECB	5A						Decrement B
DES	34						Decrement stack pointer
DEX	09						Decrement X
EORA		88	98	A8	B8		Exclusive OR into A
EORB		C8	D8	E8	F8		Exclusive OR into B
INC				6C	7C		Increment memory
INCA	4C						Increment A
INCB	5C						Increment B
INS	31						Increment stack pointer
INX	08						Increment X
JMP				6E	7E		Jump
JSR			9D	AD	BD		Jump to subroutine
LDAA		86	96	A6	B6		Load A
LDAB		C6	D6	E6	F6		Load B
LDD		CC	DC	EC	FC		Load D
LDS		8E	9E	AE	BE		Load stack pointer
LDX		CE	DE	EE	FE		Load X
LSL				68	78		Logical shift left memory (= ASL)
LSLA	48						Logical shift left A (= ASLA)
LSLB	58						Logical shift left B (= ASLB)
LSLD	05						Logical shift left D (= ASLD)
LSR				64	74		Logical shift right memory
LSRA	44						Logical shift right A
LSRB	54						Logical shift right B
LSRD	04						Logical shift right D

Mnemonic	INH	IMM	DIR	IDX	EXT	REL	Operation
MUL	3D						Multiply A by B into D (unsigned)
NEG				60	70		Negate memory
NEGA	40						Negate A
NEGB	50						Negate B
NOP	01						No operation
ORAA		8A	9A	AA	BA		OR into A
ORAB		CA	DA	EA	FA		OR into B
PSHA	36						Push A
PSHB	37						Push B
PSHX	3C						Push X
PULA	32						Pull A
PULB	33						Pull B
PULX	38						Pull X
ROL				69	79		Rotate left memory
ROLA	49						Rotate left A
ROLB	59						Rotate left B
ROR				66	76		Rotate right memory
RORA	46						Rotate right A
RORB	56						Rotate right B
RTI	3B						Return from interrupt
RTS	39						Return from subroutine
SBA	10						Subtract B from A
SBCA		82	92	A2	B2		Subtract with carry from A
SBCB		C2	D2	E2	F2		Subtract with carry from B
SEC	0D						Set carry
SEI	0F						Set interrupt mask
SEV	0B						Set overflow
STAA			97	A7	B7		Store A
STAB			D7	E7	F7		Store B
STD			DD	ED	FD		Store D
STS			9F	AF	BF		Store stack pointer
STX			DF	EF	FF		Store X
SUBA		80	90	A0	B0		Subtract from A
SUBB		C0	D0	E0	F0		Subtract from B
SUBD		83	93	A3	B3		Subtract from D
SWI	3F						Software interrupt
TAB	16						Transfer A to B
TAP	06						Transfer A to condition codes
TBA	17						Transfer B to A
TPA	07						Transfer condition codes to A
TST				6D	7D		Test memory
TSTA	4D						Test A
TSTB	5D						Test B
TSX	30						Transfer stack pointer + 1 to X
TXS	35						Transfer X - 1 to stack pointer
WAI	3E						Wait for interrupt

Appendix B

File Formats

B.1 Object files

`as01` writes object files in the Introl linkable binary format, described in the *Introl Linker and Loader Reference Manual*. They start with the two bytes `S1` (`$53 $31`).

B.2 Executables

`ld01` writes a.out executables. All multi-byte fields are big-endian. The file starts with a 32-byte header:

Offset	Size	Field	Contents
0	2	magic	`$01 $03`
2	1	flags	`$10` executable; `$20` separate instruction and data (`-i`)
3	1	cpu	`$1F` (MC6801)
4	1	header length	`$20` (32)
5	1	unused	0
6	2	version	0
8	4	text	Size of the text segment
12	4	data	Size of the data segment
16	4	bss	Size of uninitialized data
20	4	entry	Entry point address
24	4	total	Total memory used
28	4	syms	Size of the symbol table

The header is followed by the text segment, the data segment and the symbol table. Each symbol table entry is 16 bytes: an 8-byte name (padded with zeros, not necessarily terminated), a 4-byte value, a 1-byte storage class, a 1-byte auxiliary count and a 2-byte type.

`ld01 -d` writes only the text and data segments, with no header or symbols.

Appendix C

Command Reference

This appendix contains the manual pages for the tools, as installed in `doc/man`.

C.1 as01(1)

NAME

as01 - assembler for the Motorola MC6801/MC6803

SYNOPSIS

as01 [**-gkuvw**] [**-b** *binfile*] [**-l** [*listfile*]] [**-m** [*listfile*]] [**-n** *name*] [**-o** *objfile*] [**-s** *symfile*] [**-t** *n*] [*srcfile*]

DESCRIPTION

as01 assembles one MC6801 source file. It writes a relocatable object file in Introl linkable binary format for **ld01**(1), an absolute binary image, or both. It can also write a listing and a symbol table.

The assembler makes two passes over the source. If *srcfile* is omitted, it reads standard input. Options and the source file may appear in any order. Each option must be a separate argument, and each output file may be given only once.

If there are errors, the object and binary files are deleted (unless **-k** is given) and the exit status is 1.

OPTIONS

-b *binfile* Write an absolute binary image to *binfile*. Use **ORG** to set the load address. The image starts at the lowest address assembled and ends at the highest; any gaps between **ORG** areas are filled with zeros. **IMPORT**, **.GLOBL** and other external references are not allowed in this mode.

-g Put only global (exported and imported) symbols in the object file.

-k Keep the object and binary files even if there are errors.

-l [***listfile***] Write a listing to *listfile*, or to standard output if no file name follows.

-m [***listfile***] List every line of each macro expansion, including directives and lines skipped by conditionals. On its own, **-m** lists only the macro expansions; use **-l -m** *listfile* to get the full listing as well.

-n ***name*** Use *name* to derive the module name stored in the object file. By default the source file name is used. Any directory part and the last suffix are removed.

-o ***objfile*** Write a relocatable object file to *objfile*.

-s ***symfile*** Write a symbol table to *symfile*.

-t ***n*** Assemble **.TEXT** sections into location counter n+3 instead of location counter 0.

-u Treat undefined symbols as imported, as if each had been named in an **IMPORT** directive.

-w Disable warnings. Warnings are on by default (since version 2.0); **-w-** turns them on again.

-v Print a version message and exit.

The flags **-g**, **-u** and **-w** are turned off by appending **-**, for example **-w-**.

SOURCE FORMAT

Each line has the form

[*label*[**:**]] [*operation* [*operands*]] [**;** *comment*]

A label without a colon must start in column 1. A label followed by a colon may be indented. An operation must be preceded by white space unless it follows a label. A line whose first character is *****, **;** or **!** is a comment. A **;** or **!** starts a comment anywhere else on the line.

Symbols may contain letters, digits, __ and **.**, and must not start with a digit. They are case sensitive. Mnemonics, directives and register names are recognized in all upper case or all lower case.

Constants

123	decimal
$7F, **0x7F**	hexadecimal
0177	octal (leading zero)
%1010	binary
'A or **'A'**	character

Expressions

From lowest to highest precedence:

= < >	comparison (true is -1, false is 0)
+ - |	add, subtract, bitwise or
* / & « »	multiply, divide, and, shifts
- + ~	unary minus, plus, complement

Parentheses group subexpressions. A ***** in operand position is the current location counter.

Addressing modes

Inherent	**inx**	
Immediate	**ldaa #$10**	# prefix
Direct	**ldaa $80**	address in $00-$FF, chosen automatically
Extended	**ldaa $1234**	
Indexed	**ldaa 5,x**	unsigned 8-bit offset; **,x** means **0,x**
Relative	**bne loop**	branches only

A < before an operand forces direct addressing and a > forces extended. A forced direct address must fit in a byte: the assembler reports "address out of bounds" for an absolute address, or one in the module, that doesn't, and **ld01**(1) checks imported and relocated addresses when it links. Direct addressing is chosen whenever the address is in page zero, even for a symbol defined later: the assembler makes at least three passes, and more while the sizes of instructions are still changing.

A relocatable address, such as a variable that a linker script places, uses extended addressing unless its location counter was declared with **DPAGE**. An operand that is relocatable in such a counter (a symbol defined in it, imported while it was current, or declared **COMM** in it) uses direct addressing, with a one-byte address that **ld01**(1) checks is in page zero. Its address can also be used where a byte is expected, in 8-bit immediates (**ldaa #count**) and **FCB** data. This works for a symbol defined, imported or declared later in the module too.

The read-modify-write instructions (**NEG**, **COM**, **LSR**, **ROR**, **ASR**, **ASL**/**LSL**, **ROL**, **DEC**, **INC**, **TST**, **CLR**) and **JMP** have no direct mode on the MC6801 and always use extended for non-indexed operands.

DIRECTIVES

Alternative spellings are separated by /.

Data

FCB/.BYTE/DB/.DATA1 ***expr***[**,*expr...***] Define bytes. Each value is absolute, or the address of a page-zero symbol (see **DPAGE**).

FDB/.WORD/DW/.SHORT/.DATA2 ***expr*[,*expr...*]** Define 16-bit words (big-endian).

FCC/.ASCII ***delim text delim*** Define a string. The first character is the delimiter. C escape sequences such as **\n** and **\0** are recognized.

.ASCIZ ***delim text delim*** Like **FCC**, followed by a zero byte.

RMB/.BLKB/.SPACE ***count*[,*fill*]** Reserve *count* bytes, filled with *fill* if it is given.

.BLKW/.ZEROW ***count*** Reserve *count* words.

Symbols

label **EQU** ***expr*or** ***label*** **=** ***expr*** Define a constant.

label **SET** ***expr*** Define a symbol that may be redefined.

EXPORT/PUBLIC/.DEFINE ***sym*[,*sym...*]** Make symbols defined in this module visible to other modules.

ENTRY ***sym*** Like **EXPORT**, and also mark *sym* as an entry point. The linker uses the first entry point as the program's start address.

IMPORT/EXTERN/.EXTERN/EXTRN ***sym*[,*sym...*]** Declare symbols defined in another module.

.GLOBL/.GLOBAL ***sym*[,*sym...*]** Export the symbols if they are defined in this module, otherwise import them.

label **COMM** ***size*or** ***label*** **LCOMM** ***size*** Define a common block of *size* bytes. **LCOMM** makes it local to the module.

.COMM/.LCOMM ***name*,*size*** The same, in Unix assembler syntax.

Location

LOC ***n*** Switch to location counter *n* (0 to 15). The linker places counters 0 and 1 in the text segment and the others in the data segment.

DPAGE ***n*** Declare that location counter *n* (0 to 15) holds page-zero data (**$00-$FF**), from here to the end of the module. References to its symbols then use direct addressing (see **Addressing modes**). The linker script must place the counter, and its common blocks, in page zero. The modules that define and import a counter's symbols must agree on the counter, and declare it before using its symbols.

.TEXT, .DATA, .ROM, .BSS Switch to location counter 0 (**.TEXT**), 3 (**.DATA**, **.ROM**) or 4 (**.BSS**). **.BSS** is for uninitialized data, reserved with **RMB**: the default layout of **ld01**(1) gives it addresses but leaves it out of the executable, so a program that needs it to be zero clears it at start-up.

.SECT ***name*** Switch to the named section (**.text**, **.data** or **.bss**).

ORG *expr* Set the location counter. Only for an absolute program (**-b**): in a relocatable module, which the linker places, it is an error.

BLOCK *expr and* ENDB Assemble the lines in between at *expr*, then restore the previous location counter. Like **ORG**, only for an absolute program.

.ALIGN *n* Advance the location counter to a multiple of *n*.

.EVEN Advance the location counter to an even address.

Conditional assembly

IF *expr* / ELSEIF *expr* / ELSE / ENDIF Assemble lines selectively; a non-zero expression is true.

IFC *(str1,str2)* / ELSEIFC *(str1,str2)* True if the two strings are identical.

Macros

MACRO *name ...* MEND Define a macro. In the body, **?1**, **?2**, ... are replaced by the comma-separated arguments of the invocation. Invoke it as *name arg1*,*arg2*...

Other

INCLUDE/GET *file* Assemble the contents of *file*.

END End of source; the rest of the file is ignored.

FAIL Report an error.

.LIST, .MACLIST [***n***] Adjust the listing and macro-listing counters. They only have an effect when listing is enabled with **-l** or **-m**.

.WARN [***n***] Add *n* (default -1) to the warning counter. Warnings are shown while the counter is 0 or more. It starts at 0, or at -1 with **-w**, so **.WARN** turns warnings off and **.WARN 1** turns them on again.

FILES

objfile Introl linkable binary object (magic bytes **S1**).

EXAMPLES

Assemble a module and produce a listing on standard output:

```
as01 -o screen.o -l screen.s
```

Assemble an absolute image with no linking step:

```
as01 -b prog.bin prog.s
```

DIAGNOSTICS

Errors and warnings are shown in the listing under the offending line, with a caret marking the position. A summary of the error and warning counts is printed at the end. If no listing file was requested, the offending lines and the summary are printed on standard output.

SEE ALSO

ld01(1), **ar01**(1), **objdump01**(1), **objcopy01**(1)

The mc6801-tools User's Manual (*mc6801-tools-manual.pdf*).

HISTORY

Derived from the MC6809 configuration of **as09** in Bruce Evans' C compiler, as maintained in the dev86 package.

C.2 ld01(1)

NAME

ld01 - linker for MC6801 object files

SYNOPSIS

ld01 [**-dimMPrstyZ**] [**-o** *outfile*] [**-S** *script*] [**-A** *symbol=value*] [**-T** *textaddr*] [**-D** *dataaddr*] [**-H** *heapsize*] [**-C***crt*] [**-L***libdir*] [**-l***lib*] [**-O***libfile*] *file ...*

DESCRIPTION

ld01 combines object files produced by **as01**(1) and modules from archives made with **ar01**(1) into an executable. It resolves references between modules and relocates code and data to their final addresses.

Each *file* is an object file or an archive. A module is linked only if the program uses it: starting from the modules that define the entry point, **__main** or **start**, the linker includes every module that defines a symbol a linked module refers to. A common block that another module has too is no reason to include that module; the common block's storage comes with a linked module that has one, as large as the largest of any module's. Archives may appear anywhere on the command line.

A linker script controls where everything goes: which parts of which modules go into each output section, the section addresses, and extra symbols. Without **-S**, the linker uses a built-in default script, which gives the layout described under SEGMENTS. **-P** prints it, as a starting point for your own. See LINKER SCRIPTS.

The program starts at the first symbol declared with the assembler's **ENTRY** directive. Linking fails with “no start symbol” if there is none.

By default the output is an a.out executable with a 32-byte header, followed by the text segment, the data segment and a symbol table. Use **-d** for a plain memory image instead. **objcopy01**(1) converts a.out files to S-record, Intel hex, cassette and other formats.

OPTIONS

Single-letter flags may be turned off again by appending **-**, for example **-s-**.

-o *outfile* Write the output to *outfile* instead of **a.out**.

-T *textaddr* Set the address of the text segment. The value is read like a C number: hexadecimal with a **0x** or **$** prefix, octal with a leading **0**, and decimal otherwise, so **-T0x1000**, **-T'$1000'** and **-T4096** all mean $1000. (Quote **$** from the shell.) Anything else, such as **1000x**, or a value above $FFFF, is an error. The default is 0.

-D *dataaddr* Set the address of the data segment, written as for **-T**. Unless **-d** is given, this implies **-i**. By default the data segment directly follows the text segment.

-H *heapsize* Set the symbol **___heap_top** to the end of the program plus *heapsize* (written as for **-T**).

-C*crt* Link **crt***crt*.**o** as a start-up module, for example **-Cstart** for **crtstart.o**.

-l*lib* Search the archive **lib***lib*.**a.**

-L*libdir* Add *libdir* to the directories searched for **-C**, **-l** and **-O** files.

-O*libfile* Link *libfile*, looking for it in the search directories.

-d Write a headerless memory image: the text segment followed by the data segment, with no header or symbols. Implies **-s**.

-i Mark the executable as having separate instruction and data spaces.

-m List the modules linked on standard output. Each line shows the module name and its source file. + marks a linked module and **-** an archive member that was not needed.

-M Print the layout, each output section with its address and size and the input sections in it, then all symbols linked, with module, value and type, on standard output.

-S *script* Use the linker script in *script* instead of the default script.

-A *symbol=value* Define *symbol*, as if the script began with *symbol* = *value*;. The script can use it, and it takes the place of the script's **PROVIDE** for the same symbol.

-P Print the default linker script on standard output and exit. The script includes the values of **-T**, **-D**, **-H** and **-d**. Linking with it (**-S**) gives the same output as linking without a script. No input files are needed.

-Z Make it an error for the script not to place some input (see Orphans under LINKER SCRIPTS).

-r Instead of linking, write the input objects to *outfile* as an **ar01**(1) archive.

-s Do not include a symbol table in the output.

-t Trace the modules as they are read. Must come before the input files.

-y Write an extended symbol table, in which a name longer than 8 characters continues in extra entries. This is the default; **-y-** writes the old table, with names cut to 8 characters.

-v Print a version message and exit.

LIBRARY SEARCH

Files named by **-C**, **-l** and **-O** are searched for in the **-L** directories, most recently added first. There is no default library directory. If the file is not found in any of them, the name is opened as given, relative to the current directory. Up to five **-L** directories may be given.

SYMBOLS

The linker defines these symbols, which programs may import:

___btext, ___etext Start and end of the text segment.

___bdata, ___edata Start and end of the data segment.

___end End of all data, including the uninitialized data (**.BSS** and the common blocks).

___bss_start, ___bss_end Start and end of the uninitialized data, for start-up code that clears it (as for any output section, if a module imports them).

___heap_top End of the program plus the **-H** value.

___btext, **___etext**, **___bdata**, **___edata** and **___end** are always written to the executable's symbol table, unless **-s** is given. **objcopy01**(1) uses them to find the load addresses.

LINKER SCRIPTS

A script is a list of commands. Comments are written /* **...** */ or start with #. For example:

```
ENTRY(reset)

SECTIONS
{
    .text 0xE000 :
    {
        crt0.o(LOC0)
        *(LOC0 LOC1)
        *(COMMON0 COMMON1)
    }
    .data 0x0080 :
```

```
        {
            *(LOC*)
            *(COMMON)
        }
        __stack_top = 0x00FF;
    }
```

Commands

ENTRY(*symbol*) Start the program at *symbol*, before any **ENTRY** in the object files.

OUTPUT_FORMAT(*format*) **aout** (the default); **binary**, a memory image as with **-d**, covering the load addresses from the lowest to the highest used; **srec**, Motorola S-records; or **ihex**, Intel hex. S-records and Intel hex have one run of records for each section written to the output, at its load address, without the gaps between sections, and the entry point in the end record. The S-record header holds the output file's name.

MEMORY *{* name *[(*attributes*)]* : ORIGIN = *expr*, LENGTH = *expr ...* *}* Define memory regions, such as ROM and RAM. The attributes, such as **rx**, are only shown in the **-M** layout.

FILL(*byte*) The byte for gaps, where a section doesn't set its own (default 0).

ASSERT(*expression*, "*message*") Stop with *message* if *expression* is 0 once the layout is complete. The modules that define its symbols are linked.

INCLUDE "*file*" Read another script here (also allowed inside **SECTIONS**).

SECTIONS *{ ... }* Output sections and symbol assignments, in order.

***symbol* = *expression*;** Define a symbol.

PROVIDE(*symbol* = *expression*); Define it only if a linked module imports it and no module defines it.

HIDDEN(*symbol* = *expression*); Define a symbol that only the script can use.

Output sections

```
name [address] [(NOLOAD)] : [AT(lma)] [ALIGN(n)] [SUBALIGN(n)]
    { statements } [> region] [AT> region] [= fill]
```

Without an *address*, a section placed in a region (*> region*) starts at the region's next free address, and other sections start at the current address (**.**), which is where the previous section ended. It's an error if a section doesn't fit in its region.

A section is stored at its load address, which is the same as its address unless **AT** gives another one, or **AT>** allocates it from a region. For example, initialized data that runs in RAM can be stored in ROM after the code, and copied by the start-up code. Separate load addresses need the **binary**, **srec** or **ihex** output format.

NOLOAD sections take up addresses but aren't written to the output. They may only hold space reserved with **RMB** and common blocks: a module's initialized data, or **BYTE** or **WORD**, in one is an error, since it would be lost. **= fill** sets the byte for gaps in the section. **ALIGN** rounds the start up to a multiple of *n*. **SUBALIGN** pads each input section in it to a multiple of *n* bytes.

Statements inside a section, in order:

***file*(*sections*)** Add the matching input sections.

ONEPAGE *{ statements }* Keep the contents within one 256-byte page: if they would end on a later page than they start, they move to the start of the next page.

symbol* = *expression*;, *PROVIDE*, *HIDDEN Define symbols, often using **.**, the current address.

. = *expression*; Move the current address forward. The gap is filled with the fill byte.

KEEP(*file*(*sections*)) Add input sections like *file*(*sections*)**,** and link their modules even if nothing refers to them.

BYTE(*expression*); WORD(*expression*); Put a byte, or a 16-bit word (high byte first), into the section.

FILL(*byte*); Use *byte* for the gaps that follow.

ASSERT(*expression*, "*message*"); As at the top level, but with **.** as here.

Input sections

Each linked module has two input sections for each location counter *n*: **LOC***n,* its code and data, and **COMMON***n,* its common blocks. **.text** means **LOC0**, **.data** means **LOC3**, **.bss** means **LOC4**, and **COMMON** means all the common blocks.

file matches the object file name, with or without its directory; archive members are named *archive***:***member*. Both *file* and the section names may use the wildcards ***** and **?**. Each input section goes to the first statement that matches it. A statement adds its input sections in module order, and each module's in counter order.

A statement can start with **EXCLUDE_FILE(***file ...***),** naming files (patterns, as above) that it doesn't take input from, also inside **KEEP**: **EXCLUDE_FILE(vectors.o) *(LOC0 LOC1)** takes counters 0 and 1 of every module except **vectors.o**, so that a later section can place it.

Orphans

Input that no statement matches is added, with a warning, at the end of the last output section for its kind (counters 0 and 1, or the others) that takes input from all files (*****, without **EXCLUDE_FILE**), or else the last section for its kind. **-Z** makes it an error.

Expressions

Expressions use C operators and numbers written as for **-T**. A **K** suffix multiplies by 1024. The functions are **ADDR(***section***), SIZEOF(***section***), LOADADDR(***section***), ALIGN(***n***)** (the current address rounded up), **ALIGN(***expression***,** *n***), DEFINED(***symbol***), ORIGIN(***region***), LENGTH(***region***), MAX** and **MIN**. Symbols may be used before they're defined; the linker repeats the layout until nothing changes.

Symbols

Symbols defined by a script are absolute. They satisfy imports from any location counter. A plain assignment to a symbol that a module also defines is an error.

For each output section, the linker also provides **___***name***__start,** **___***name***__end** and **___***name***__load** (its load address), with the section name's leading dot dropped, if a module imports them. For example, start-up code can copy **.data** from **___data__load** to **___data__start**.

a.out output

The output sections up to the first one that can take input from location counters 2 to 15 form the a.out text part, and the rest the data part. The output is marked as having separate instruction and data spaces if the data part doesn't start where the text part ends. Uninitialized data at the end of the data part, such as a **NOLOAD** section, is the a.out header's bss: it isn't written to the file. The header's total memory size reaches **___heap__top** if the script sets it.

SEGMENTS

This is the layout of the default script. The assembler has 16 location counters. Counters 0 and 1 are linked into the text segment, and counters 2, 3 and 5 through 15 into the data segment. Counter 4 and all common blocks are the uninitialized data, after the data: a **NOLOAD** section, **.bss**, which isn't written to the file. Each module's part of a segment follows the one before it, with no padding. If counter 2 would cross a 256-byte page boundary, it is moved to the start of the next page. **.TEXT** selects counter 0, **.DATA** counter 3, and **.BSS** counter 4.

Before version 2.0, **.BSS** was counter 3, each module was padded to a multiple of 4 bytes, and common blocks were rounded up to 4 bytes; the default script that **ld01 -P** printed then still gives that layout.

EXAMPLES

Link a start-up module, two objects and a library, loading at $434C:

```
ld01 -Cstart -lnum -T0x434c -o main.aout main.o screen.o
```

Produce a raw memory image for an EPROM at $E000:

ld01 -d -T0xe000 -o rom.bin start.o prog.o

Start a linker script from the default layout, edit it, and use it:

ld01 -P -T0x434c > mc10.ld ld01 -S mc10.ld -Cstart -lnum -o main.aout main.o screen.o

DIAGNOSTICS

Undefined symbols are reported as "undefined symbol: *name*".

A symbol imported under a different location counter from the one it is defined in is reported with a warning, such as "counter is imported under LOC0 in main.o, but defined in LOC3 in vars.o". The link uses the definition, but the importing module chose its addressing for the counter it was imported under (see **DPAGE** in **as01**(1)).

A one-byte address, from a direct address forced with < or an indexed offset, must end up in page zero (**$00-$ff**). One that doesn't is an error, naming the module (or library member), the symbol, or the location counter and offset in the module, its address, and the address of the byte:

ld01: main.o: count ($5004) is not in page zero (00−ff), at $e001 ld01: main.o: LOC3+$0002 ($5002) is not in page zero (00−ff), at $e005

Initialized data in a module's location counter that the script puts in a **NOLOAD** section is an error, reported once for each module and counter, as in "main.o: LOC4 has initialized data, but .bss is NOLOAD".

The exit status is 0 on success and non-zero on error.

SEE ALSO

as01(1), **ar01**(1), **objdump01**(1), **objcopy01**(1)

The mc6801-tools User's Manual (*mc6801-tools-manual.pdf*).

HISTORY

Derived from **ld86** in Bruce Evans' C compiler, as maintained in the dev86 package.

C.3 ar01(1)

NAME

ar01 - create and maintain libraries of MC6801 object files

SYNOPSIS

ar01 [**-**]*key* [*posname*] *archive* [*file...*]

DESCRIPTION

ar01 maintains archives (libraries) of object files produced by **as01**(1). **ld01**(1) searches archives with its **-l** option and links only the members that are needed. The archive format is the standard Unix **!<arch>** format.

key is one operation letter, optionally combined with modifier letters. The leading **-** is optional.

OPERATIONS

r Replace the named files in the archive, or add them at the end if they are not already there. Creates the archive if it does not exist.

q Quick append: add the named files at the end without checking for existing members.

d Delete the named members.

m Move the named members, by default to the end of the archive.

t List the members, or only the named ones.

x Extract the named members into the current directory, or all members if none are named.

p Write the contents of the named members (or all members) to standard output.

MODIFIERS

a With **r** or **m**, place the files after member *posname*.

b, i With **r** or **m**, place the files before member *posname*.

c Do not report when a new archive is created.

o On extraction, give files the modification time recorded in the archive.

u With **r**, replace only members older than the named files.

v Verbose: report each file added, replaced or extracted, and give a long listing with **t**.

l Accepted and ignored.

s Add or update a **___.SYMDEF** member, an index of the symbols that each member exports, including common blocks. Once an archive has an index, later changes to the archive keep it up to date. **ld01**(1) does not need the index; it reads every member's symbols directly. Members that are not object files are reported and left out of the index.

EXAMPLES

Create a library from three modules:

ar01 rc libnum.a one.o two.o three.o

List its contents in detail:

ar01 tv libnum.a

Link against it:

ld01 -Cstart -lnum -T0x434c -o main.aout main.o

SEE ALSO

as01(1), **ld01**(1), **objdump01**(1)
The mc6801-tools User's Manual (*mc6801-tools-manual.pdf*).

HISTORY

Derived from the GNU **ar** used in the dev86 package.

C.4 objdump01(1)

NAME

objdump01, nm01, size01 - display information about MC6801 object files and executables

SYNOPSIS

objdump01 [**-n**|**-s**] [**-d**] [**-o**] *file ...*
nm01 [**-o**] *file ...*
size01 *file ...*

DESCRIPTION

objdump01 displays the contents of object files produced by **as01**(1), archives of them made by **ar01**(1), and a.out executables produced by **ld01**(1). **make install** also installs it as **nm01** and **size01**, which behave as if **-n** or **-s** had been given. The mode is chosen from the first letter of the program name.

By default **objdump01** prints for each executable:

- the header: header length, CPU (m6801), flags, entry point (**a__entry**), total memory size (**a__total**) and symbol table size;
- the segment sizes;
- the symbol table;
- a hex and ASCII dump of the text and data segments.

For each object file, or each object file in an archive, it prints:

- the module name and the size of each location counter (**SEG0**, **SEG3**, ...);
- the symbol table, with flags for exported (**E**), imported (**I**), entry point (**N**) and common (**C**) symbols;
- the text, data and bss sizes;
- the module's code, with the relocation records that **ld01**(1) uses to fill in addresses, such as **DW clrscn** for a 2-byte reference to the symbol **clrscn**.

In object files, location counters 0 and 1 count as text, counter 4 (**.BSS**) as bss, and the others as data, as they are linked by **ld01**(1).

With **-d**, the code is also disassembled.

OPTIONS

-d After the rest of the output, disassemble the code: an executable's text segment, at the address of its **___btext** symbol, and location counters 0 and 1 of each object module, from offset 0. Each line gives the address, the instruction's bytes, a label if a symbol is defined there, the mnemonic and the operand, in **as01**(1) syntax. In object files, an operand that **ld01**(1) fills in is shown by name: the imported symbol, or the label nearest before the address in the module (such as **local+$2**), with < for a direct address and # for an immediate. In executables, addresses and branch targets are shown by name where a symbol has that value. Bytes that aren't an instruction are shown with **fcb**, a relocated field that isn't an operand with **fdb**, and an area that wasn't loaded (**RMB**) with **rmb**.

-n Print only the symbol table, like **nm**(1). Each line gives the value, a type letter and the name.

-s Print only the text, data and bss sizes, in decimal, followed by the total in decimal and hexadecimal, like **size**(1). If several files are given, a total line is added.

-o In **-n** mode, prefix each line with the file name instead of printing a heading for each file.

SYMBOL TYPES

T, **t**	text segment symbol (global, local)
D, **d**	data segment symbol (global, local)
C, **b**	bss or common symbol (global, local)
a	absolute symbol
U	undefined
?	other

EXAMPLES

```
$ size01 main.aout
text    data    bss dec hex filename
108 0   0   108 6c  main.aout

$ nm01 main.aout
0000434c t msg_one
00004351 T one
00004380 T __start
00004000 a SCREEN
...

$ nm01 main.o
00000000 T main
         U three
         U one
...
```

DIAGNOSTICS

Errors, such as a file that can't be opened or isn't an object file or executable, are reported and the next file is examined. In the default mode they are printed on standard output, as **Error:** *message,* among the rest of the output; **nm01** and **size01** print them on standard error.

The exit status is 0 if every file could be read, and 1 if there was an error or no file was given.

SEE ALSO

ld01(1), **objcopy01**(1), **as01**(1)

The mc6801-tools User's Manual (*mc6801-tools-manual.pdf*).

BUGS

The disassembler can't tell data from code: data in a code segment, such as a string, is shown as instructions where its bytes happen to be opcodes.

Unknown options are silently ignored.

HISTORY

Derived from **objdump86** in the dev86 package, by Greg Haerr and others.

C.5 objcopy01(1)

NAME

objcopy01 - convert MC6801 executables to loadable formats

SYNOPSIS

objcopy01 [**-O** *format*] [**-T** *textaddr*] [**-D** *dataaddr*] [**-n** *name*] [**-p**] *infile outfile*

DESCRIPTION

objcopy01 reads an a.out executable produced by **ld01**(1) and writes its text and data segments to *outfile* in one of several formats for loaders, EPROM programmers and cassette interfaces. The input must have an a.out header, so do not link it with **ld01 -d**.

The program's entry point is taken from the a.out header. The load addresses are taken from the **___btext** and **___bdata** symbols that **ld01**(1) writes to the symbol table. **-T** and **-D** override them. An executable linked with **ld01 -s** has no symbols, so give **-T** for it.

ld01(1) can also write S-records, Intel hex and binary images directly, with **OUTPUT_FORMAT** in a linker script. That handles programs spread over several areas of memory and separate load addresses, which a.out files can't hold.

OPTIONS

-O ***format*** Output format (default **binary**). See FORMATS.

-T ***textaddr*** Load address of the text segment. The default is the value of **___btext**. The value is read like a C number: hexadecimal with a **0x** or **$** prefix, octal with a leading **0**, and decimal otherwise, so **-T0x1000**, **-T'$1000'** and **-T4096** all mean $1000. (Quote **$** from the shell.) Anything else, such as **1000x**, or a value above $FFFF, is an error. This is the same as **ld01 -T**.

-D ***dataaddr*** Load address of the data segment, for formats that can place it separately. The default is the value of **___bdata**; if there are no symbols, the data segment follows the text segment.

-n ***name*** Name to store in the output. This is the S0 header record name for **srec**, the cassette file name for **cas**/**c10** and **wav** (up to 8 characters, padded with spaces). The default is the output file name in upper case, including any directory and suffix, so it is usually best to give **-n** explicitly.

-p With **binary**, **wav** or **cas** (formats of one block), for a program whose data isn't just after its text (above it or below it), write one image from the lower of their addresses to the end of the higher, the gap between them filled with zeros: each is then at its address's place, and the image loads at the lower address (the entry point is still the program's). Text and data that overlap, or span more than 64K, are an error.

FORMATS

The format name may be abbreviated to its first three letters.

binary Raw memory image: the text segment followed by the data segment (with **-p**, each at its place).

srec Motorola S-records: an S0 header containing the name, S1 data records of 16 bytes, an S5 record count and an S9 record with the entry point.

ihex Intel hex: 16-byte data records and an end record holding the entry point.

wav Cassette audio as an 8-bit mono WAV file at 43200 Hz, in the Color BASIC tape format. This format is also used by the Tandy MC-10, which loads it with **CLOADM**. A machine-language file is one block of memory, so data that isn't just after the text needs **-p**.

cas, c10 The same cassette data as **wav**, as a byte stream for emulators. **c10** is another name for the same format, matching the **.c10** files used by MC-10 emulators.

EXAMPLES

Make an MC-10 cassette image:

objcopy01 -O wav -n MAIN main.aout main.wav

Make an S-record file from an executable linked with **ld01 -s -T0xe000**, which has no symbols to give the load address:

objcopy01 -O srec -T 0xe000 -n ROM rom.aout rom.s19

DIAGNOSTICS

"*format* load address is zero (use -T)" means the input has no symbol table (or was linked at address 0); give **-T**. "*format* data load address is zero (use -D)" is the same for a separate data segment.

SEE ALSO

ld01(1), **objdump01**(1), **as01**(1)

The mc6801-tools User's Manual (*mc6801-tools-manual.pdf*).

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